



Computer Network Project

DESIGN AND CONFIGURATION OF A MULTI-DEPARTMENT COLLEGE COMPUTER NETWORK

Under the Guidance of

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Introduction

College Network System

A College Network System is a computer network designed to connect different departments, laboratories, administrative offices, and servers within a college campus. It enables efficient communication, resource sharing, and centralized management of network services. A well-designed campus network improves security, reliability, and performance while allowing users to access shared resources such as the internet, file servers and web servers.

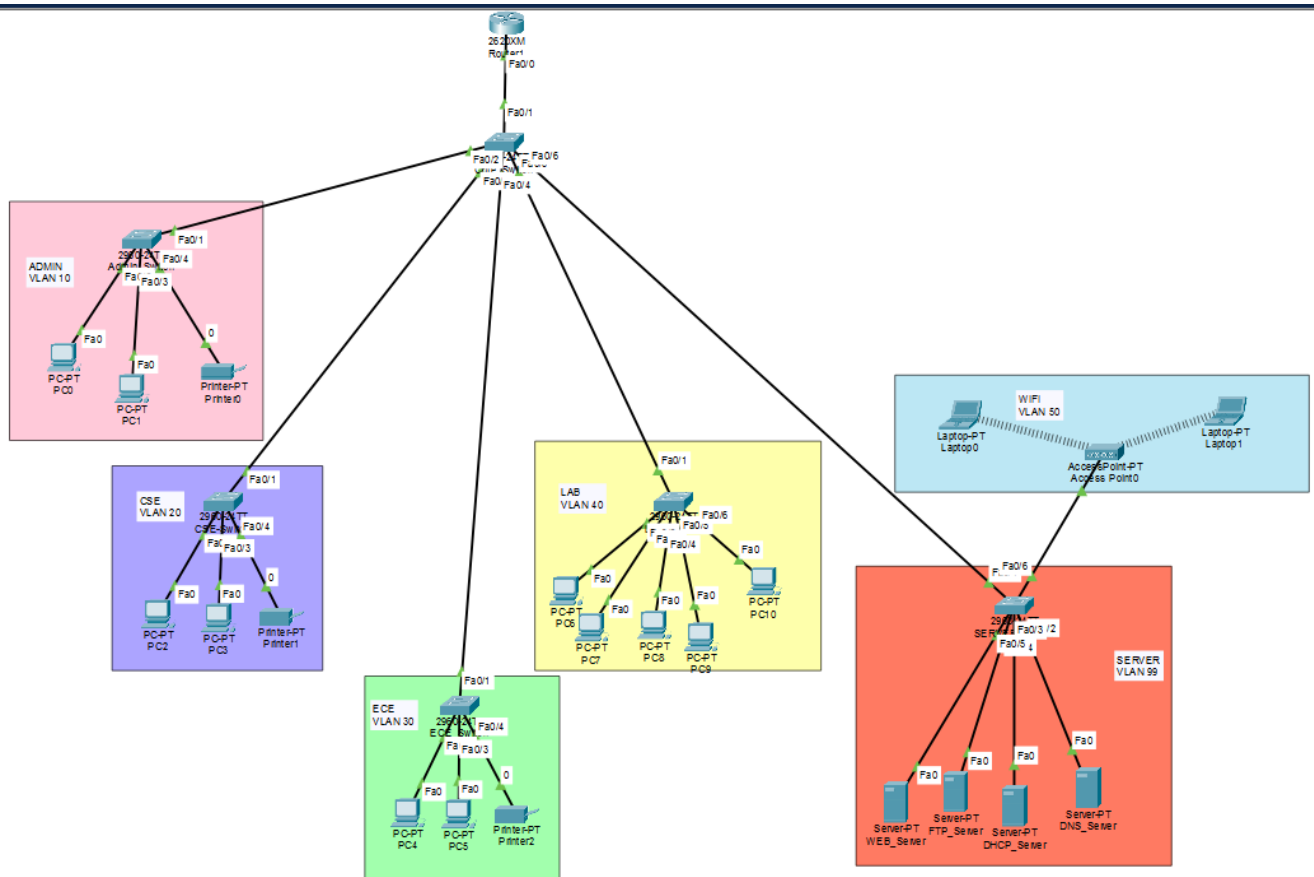
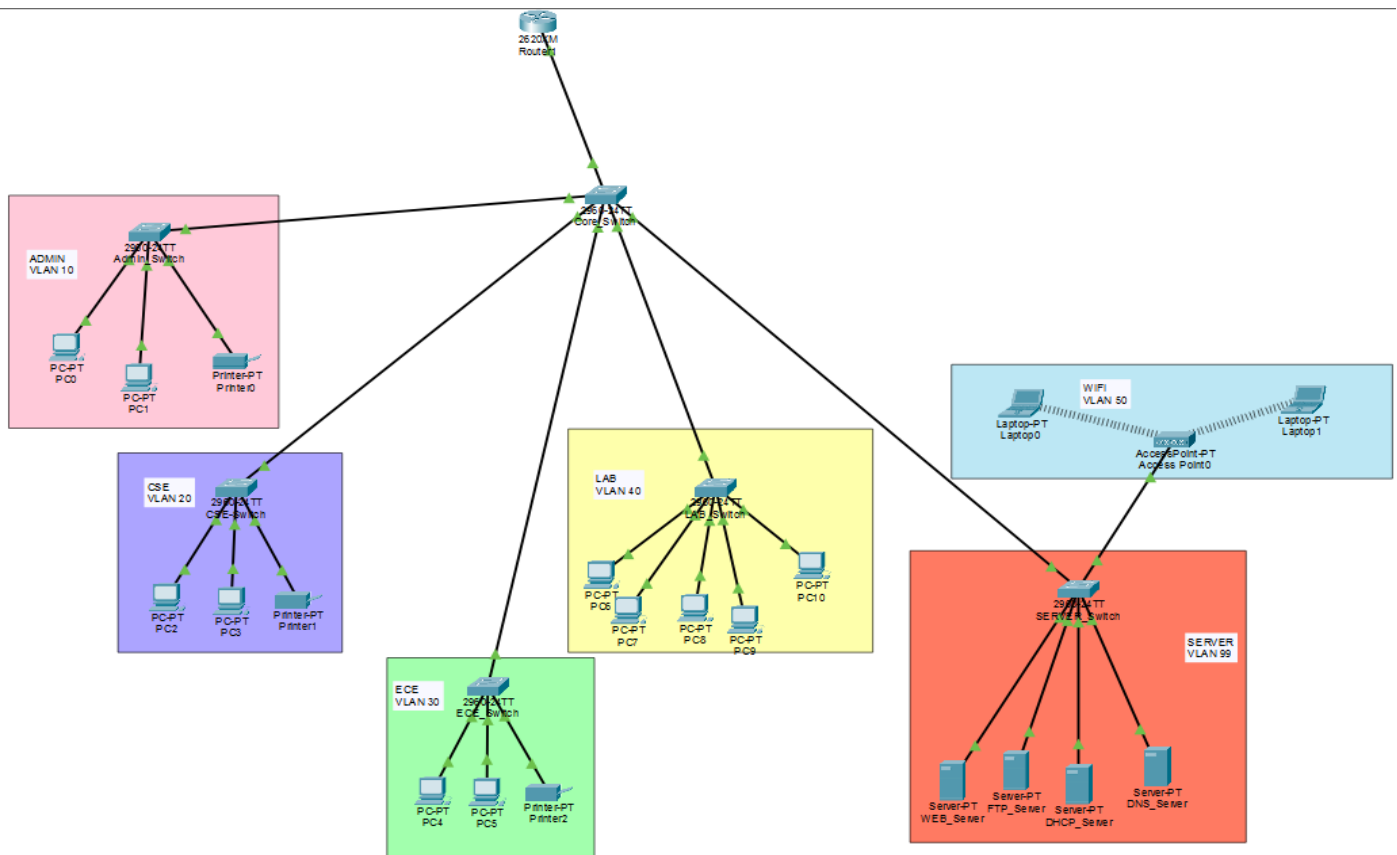
In this project, a college campus network has been designed and simulated using Cisco Packet Tracer. The network consists of multiple departments connected through switches and a router. Virtual Local Area Networks (VLANs) are used to separate different departments such as Admin, CSE, ECE, and Lab, providing better security and reducing unnecessary network traffic. Inter-VLAN routing is implemented using a router to allow communication between different VLANs.

The project also includes a DHCP server for automatic IP address assignment, a DNS server for domain name resolution, a Web server for hosting web pages, an FTP server for file sharing, and a Wireless Access Point to provide Wi-Fi connectivity to wireless devices. The objective of this project is to demonstrate the design, configuration, and implementation of a secure, scalable, and efficient college campus network.

Devices used: 1 Router (2620XM) → Router1

- 1 Core switch (2960-24TT) → Core_Switch
- 5 access switches (2960-24TT) → Admin_Switch, CSE_Switch, ECE_Switch, lab_Switch, Server_Switch
- 1 Access Point (WRT300N or AccessPoint-PT) → AccessPoint0
- PCs: PC0–PC1 (Admin), PC2–PC3(CSE), PC4–PC5(ECE), PC6–PC10 (Lab)
- Printers: Printer0 (Admin), Printer1 (CSE), Printer2 (ECE)
- Laptops: Laptop0, Laptop1 (Wi-Fi)
- Servers: DHCP_Server, DNS_Server, WEB_Server, FTP_Server

Network Diagram:



Router Configuration:

Click on Router → Select CLI → Write these commands

```
Router1
Physical Config CLI Attributes

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#interface fastEthernet0/0
Router1(config-if)#no shutdown

Router1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router1(config)#interface fastEthernet0/0.10
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router1(config-subif)#ip address 192.168.10.1 255.255.255.0
Router1(config-subif)#ip helper-address 192.168.99.20
Router1(config-subif)#exit
Router1(config)#interface fastEthernet0/0.20
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router1(config-subif)#ip address 192.168.20.1 255.255.255.0
Router1(config-subif)#ip helper-address 192.168.99.20
Router1(config-subif)#exit
Router1(config)#interface fastEthernet0/0.30
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up
encapsulation dot1Q 30
Router1(config-subif)#ip address 192.168.30.1 255.255.255.0
Router1(config-subif)#ip helper-address 192.168.99.20
Router1(config-subif)#exit
Router1(config)#interface fastEthernet0/0.40
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.40, changed state to up
encapsulation dot1Q 40
Router1(config-subif)#ip address 192.168.40.1 255.255.255.0
Router1(config-subif)#ip helper-address 192.168.99.20
Router1(config-subif)#exit
Router1(config)#interface fastEthernet0/0.50
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.50, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.50, changed state to up
encapsulation dot1Q 50
Router1(config-subif)#ip address 192.168.50.1 255.255.255.0
Router1(config-subif)#ip helper-address 192.168.99.20
Router1(config-subif)#exit
Router1(config)#interface fastEthernet0/0.99
Router1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.99, changed state to down

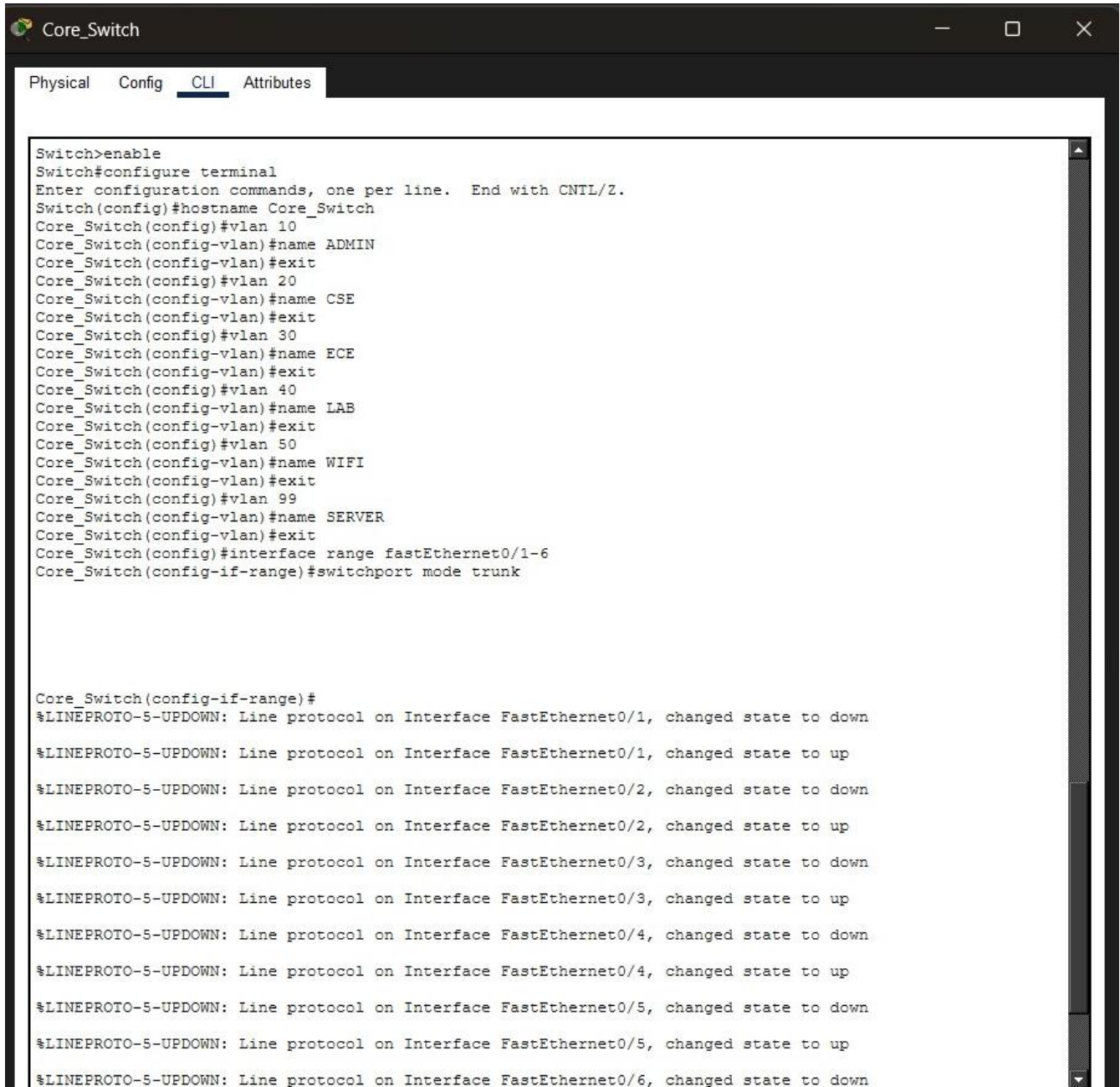
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.99, changed state to up
encapsulation dot1Q 99
Router1(config-subif)#ip address 192.168.99.1 255.255.255.0
Router1(config-subif)#exit
```

Copy Paste

Top

Switch Configuration:

Click on Switch → Select CLI → Write these commands



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Core_Switch
Core_Switch(config)#vlan 10
Core_Switch(config-vlan)#name ADMIN
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 20
Core_Switch(config-vlan)#name CSE
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 30
Core_Switch(config-vlan)#name ECE
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 40
Core_Switch(config-vlan)#name LAB
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 50
Core_Switch(config-vlan)#name WIFI
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 99
Core_Switch(config-vlan)#name SERVER
Core_Switch(config-vlan)#exit
Core_Switch(config)#interface range fastEthernet0/1-6
Core_Switch(config-if-range)#switchport mode trunk

Core_Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
```

VLAN Switch Configuration:

For ADMIN Switch

Click on Switch → Select CLI → Write these commands

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Admin_Switch
Admin_Switch(config)#vlan 10
Admin_Switch(config-vlan)#name ADMIN
Admin_Switch(config-vlan)#exit
Admin_Switch(config)#interface fastEthernet0/1
Admin_Switch(config-if)#switchport mode trunk
Admin_Switch(config-if)#no shutdown
Admin_Switch(config-if)#exit
Admin_Switch(config)#interface range fastEthernet0/2-24
Admin_Switch(config-if-range)#switchport mode access
Admin_Switch(config-if-range)#switchport access vlan 10
Admin_Switch(config-if-range)#no shutdown
Admin_Switch(config-if-range)#exit
Admin_Switch(config)#exit
Admin_Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Admin_Switch#
```

For CSE Switch

Click on Switch → Select CLI → Write these commands

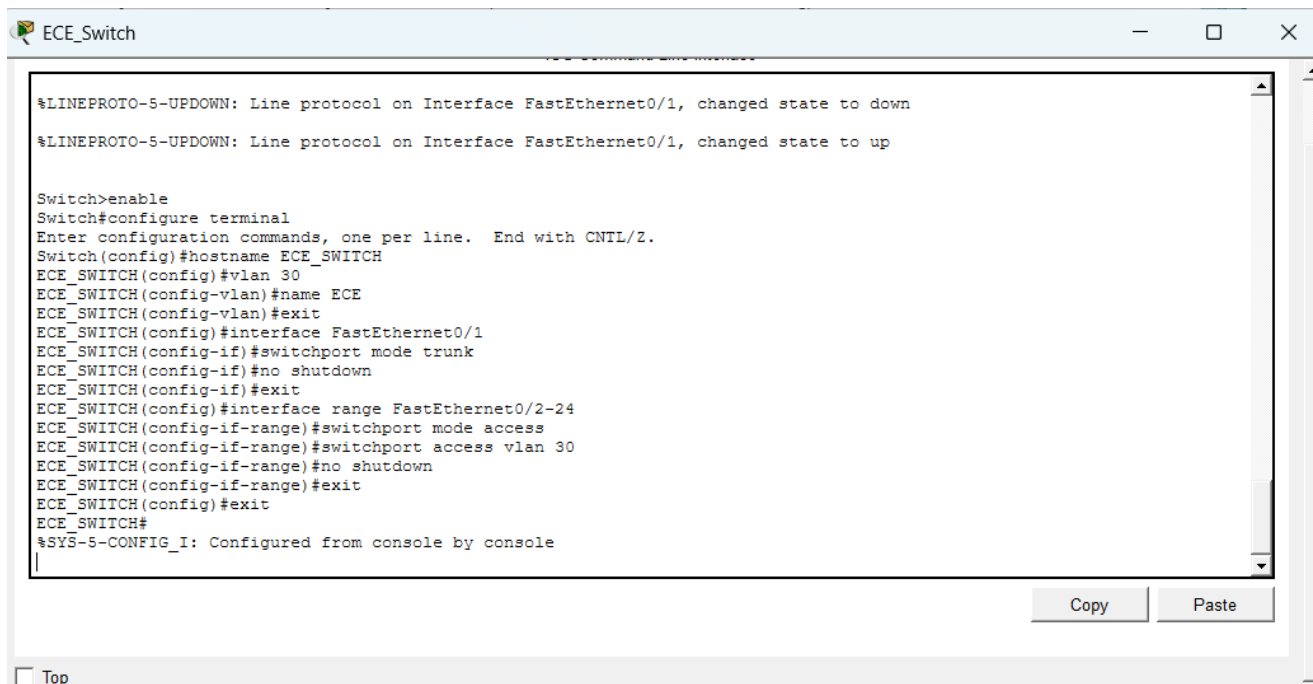
```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname CSE_Switch
CSE_Switch(config)#vlan 20
CSE_Switch(config-vlan)#name CSE
CSE_Switch(config-vlan)#exit
CSE_Switch(config)#interface fastEthernet0/1
CSE_Switch(config-if)#switchport mode trunk
CSE_Switch(config-if)#no shutdown
CSE_Switch(config-if)#exit
CSE_Switch(config)#interface range fastEthernet0/2-24
CSE_Switch(config-if-range)#switchport mode access
CSE_Switch(config-if-range)#switchport access vlan 20
CSE_Switch(config-if-range)#no shutdown
CSE_Switch(config-if-range)#exit
CSE_Switch(config)#exit
CSE_Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
CSE_Switch#
```

Copy

Paste

For ECE Switch

Click on Switch → Select CLI → Write these commands



The screenshot shows a terminal window titled "ECE_Switch". The terminal output displays the following commands and their results:

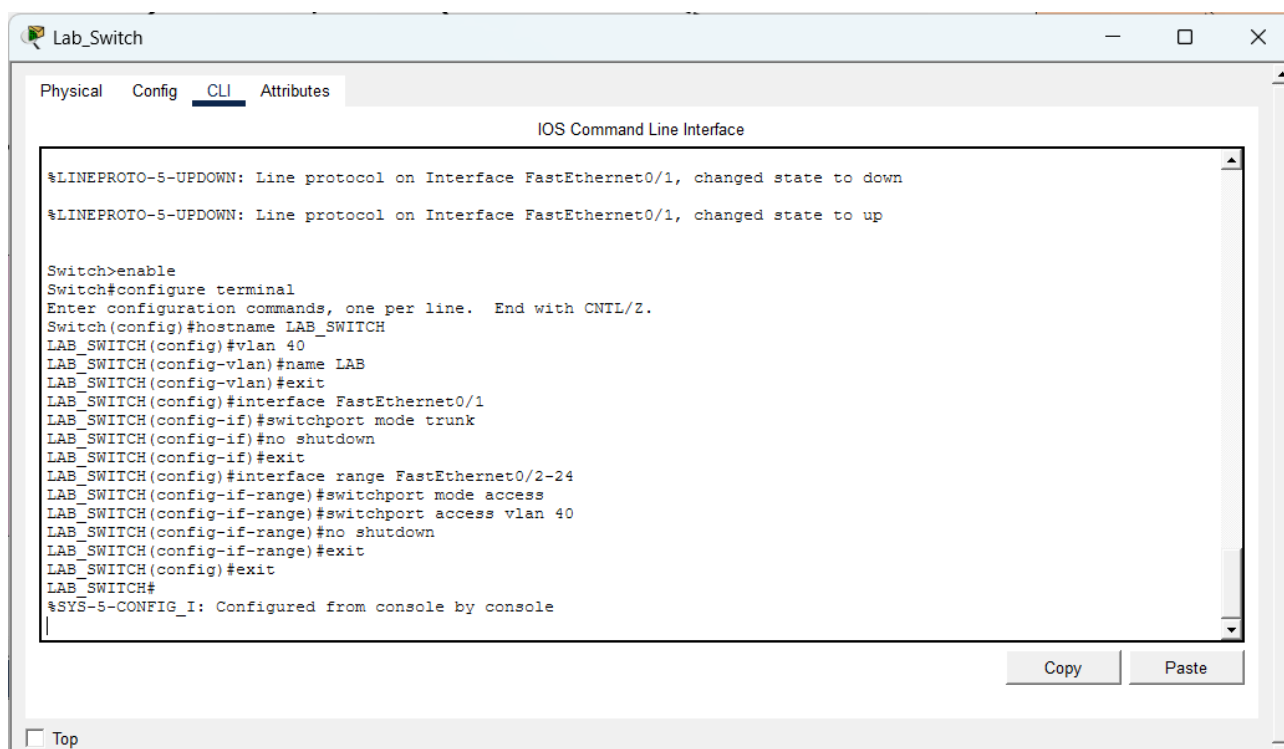
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname ECE_SWITCH
ECE_SWITCH(config)#vlan 30
ECE_SWITCH(config-vlan)#name ECE
ECE_SWITCH(config-vlan)#exit
ECE_SWITCH(config)#interface FastEthernet0/1
ECE_SWITCH(config-if)#switchport mode trunk
ECE_SWITCH(config-if)#no shutdown
ECE_SWITCH(config-if)#exit
ECE_SWITCH(config)#interface range FastEthernet0/2-24
ECE_SWITCH(config-if-range)#switchport mode access
ECE_SWITCH(config-if-range)#switchport access vlan 30
ECE_SWITCH(config-if-range)#no shutdown
ECE_SWITCH(config-if-range)#exit
ECE_SWITCH(config)#exit
ECE_SWITCH#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. A "Top" button is located at the bottom left of the window frame.

For LAB Switch

Click on Switch → Select CLI → Write these commands



The screenshot shows a terminal window titled "Lab_Switch" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, and the terminal output displays the following commands and their results:

```
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname LAB_SWITCH
LAB_SWITCH(config)#vlan 40
LAB_SWITCH(config-vlan)#name LAB
LAB_SWITCH(config-vlan)#exit
LAB_SWITCH(config)#interface FastEthernet0/1
LAB_SWITCH(config-if)#switchport mode trunk
LAB_SWITCH(config-if)#no shutdown
LAB_SWITCH(config-if)#exit
LAB_SWITCH(config)#interface range FastEthernet0/2-24
LAB_SWITCH(config-if-range)#switchport mode access
LAB_SWITCH(config-if-range)#switchport access vlan 40
LAB_SWITCH(config-if-range)#no shutdown
LAB_SWITCH(config-if-range)#exit
LAB_SWITCH(config)#exit
LAB_SWITCH#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. A "Top" button is located at the bottom left of the window frame.

For SERVER Switch

Click on Switch → Select CLI → Write these commands

The screenshot displays a network simulation environment. On the left, a network diagram shows a central switch connected to a router (2621) and several PCs (PC6, PC7). The switch has multiple ports labeled Fa0/0 through Fa0/24. The main window is titled 'Server_Switch' and has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, showing a command-line interface with the following text:

```
technical support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Server_Switch
Server_Switch(config)#vlan 99
Server_Switch(config-vlan)#name SERVER
Server_Switch(config-vlan)#exit
Server_Switch(config)#vlan 50
Server_Switch(config-vlan)#Server_Switch(config-vlan)#vlan 50
Server_Switch(config-vlan)#^
% Invalid input detected at '^' marker.

Server_Switch(config-vlan)#name WIFI
Server_Switch(config-vlan)#exit
Server_Switch(config)#interface fastEthernet0/1
Server_Switch(config-if)#switchport mode trunk
Server_Switch(config-if)#no shutdown
Server_Switch(config-if)#exit
Server_Switch(config)#interface range fastEthernet0/2-5
Server_Switch(config-if-range)#switchport mode access
Server_Switch(config-if-range)#switchport access vlan 99
Server_Switch(config-if-range)#no shutdown
Server_Switch(config-if-range)#exit
Server_Switch(config)#interface fastEthernet0/6
Server_Switch(config-if)#switchport mode access
Server_Switch(config-if)#switchport access vlan 50
Server_Switch(config-if)#no shutdown
Server_Switch(config-if)#exit
Server_Switch(config)#exit
Server_Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Server_Switch#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. The simulation interface also includes a 'Simulation' button on the right side.

Server IP Configuration:

DNS SERVER

Click on DNS Server → Select Desktop → Click on IP Configuration
→ Enter IP Address and Subnet Mask

The screenshot shows the 'DNS_Server' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the following settings:

Field	Value
IP Configuration	☒ Static
IPv4 Address	192.168.99.21
Subnet Mask	255.255.255.0
Default Gateway	192.168.99.1
DNS Server	0.0.0.0

Below the IPv4 section, the IPv6 Configuration section is visible with the following settings:

Field	Value
IPv6 Configuration	☒ Static
IPv6 Address	
Link Local Address	FE80::20A:41FF:FE14:6597
Default Gateway	
DNS Server	

The 802.1X section is also visible with the following settings:

Field	Value
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	

DHCP Server

Click on DHCP Server → Select Desktop → Click on IP Configuration
Enter IP Address and Subnet Mask

The screenshot shows the 'DHCP_Server' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the following settings:

Field	Value
IP Configuration	☒ Static
IPv4 Address	192.168.99.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.99.1
DNS Server	0.0.0.0

Below the IPv4 section, the IPv6 Configuration section is visible with the following settings:

Field	Value
IPv6 Configuration	☒ Static
IPv6 Address	
Link Local Address	FE80::2D0:BCFF:FE33:7446
Default Gateway	
DNS Server	

The 802.1X section is also visible with the following settings:

Field	Value
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	

WEB Server

Click on WEB Server → Select Desktop → Click on IP Configuration
Enter IP Address and Subnet Mask

The screenshot shows the 'WEB_Server' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing settings for both IPv4 and IPv6. The IPv4 configuration is set to 'Static' with an IP address of 192.168.99.22, a subnet mask of 255.255.255.0, a default gateway of 192.168.99.1, and a DNS server of 192.168.99.21. The IPv6 configuration is also set to 'Static' with a link local address of FE80::206:2AFF:FE57:3E96. The 802.1X section is visible at the bottom, with 'Use 802.1X Security' unchecked and 'Authentication' set to 'MD5'.

IP Configuration	
☐ DHCP ☒ Static	
IPv4 Address	192.168.99.22
Subnet Mask	255.255.255.0
Default Gateway	192.168.99.1
DNS Server	192.168.99.21

IPv6 Configuration	
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::206:2AFF:FE57:3E96
Default Gateway	
DNS Server	

802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

FTP Server

Click on FTP Server → Select Desktop → Click on IP Configuration
Enter IP Address and Subnet Mask

The screenshot shows the 'FTP_Server' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing settings for both IPv4 and IPv6. The IPv4 configuration is set to 'Static' with an IP address of 192.168.99.23, a subnet mask of 255.255.255.0, a default gateway of 192.168.99.1, and a DNS server of 0.0.0.0. The IPv6 configuration is also set to 'Static' with a link local address of FE80::201:97FF:FE06:DC19. The 802.1X section is visible at the bottom, with 'Use 802.1X Security' unchecked and 'Authentication' set to 'MD5'.

IP Configuration	
☐ DHCP ☒ Static	
IPv4 Address	192.168.99.23
Subnet Mask	255.255.255.0
Default Gateway	192.168.99.1
DNS Server	0.0.0.0

IPv6 Configuration	
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::201:97FF:FE06:DC19
Default Gateway	
DNS Server	

802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

DHCP Server Configuration:

Click on DHCP Server → Select Services → Go to DHCP

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: ADMIN

Default Gateway: 192.168.10.1

DNS Server: 192.168.99.21

Start IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Maximum Number of Users: 200

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
WIFI	192.168.50.1	192.168.99.21	192.168.50.10	255.255.255.0	200	0.0.0.0	0.0.0.0
LAB	192.168.40.1	192.168.99.21	192.168.40.10	255.255.255.0	200	0.0.0.0	0.0.0.0
ECE	192.168.30.1	192.168.99.21	192.168.30.10	255.255.255.0	200	0.0.0.0	0.0.0.0
CSE	192.168.20.1	192.168.99.21	192.168.20.10	255.255.255.0	200	0.0.0.0	0.0.0.0
ADMIN	192.168.10.1	192.168.99.21	192.168.10.10	255.255.255.0	200	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.99.0	255.255.255.0	512	0.0.0.0	0.0.0.0

WEB Server Configuration:

Go to WEB Server → Select Services → Click on HTTP

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

HTTP

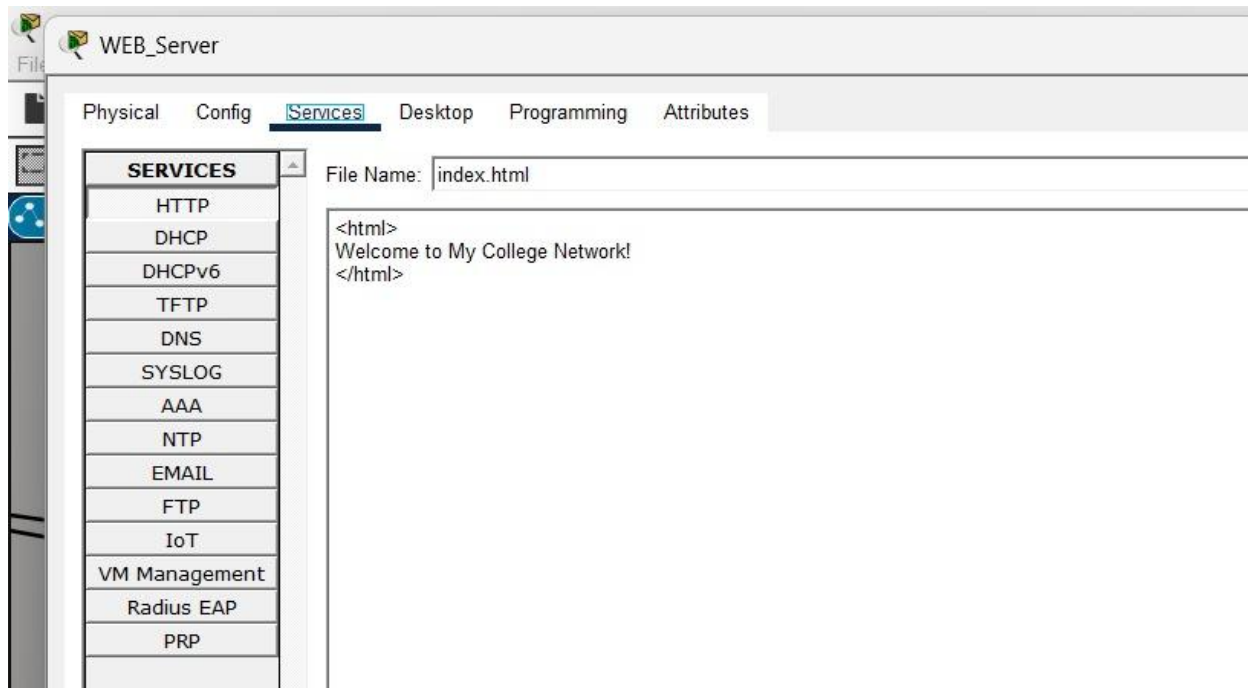
HTTP: ☒ On ☐ Off

HTTPS: ☒ On ☐ Off

File Manager

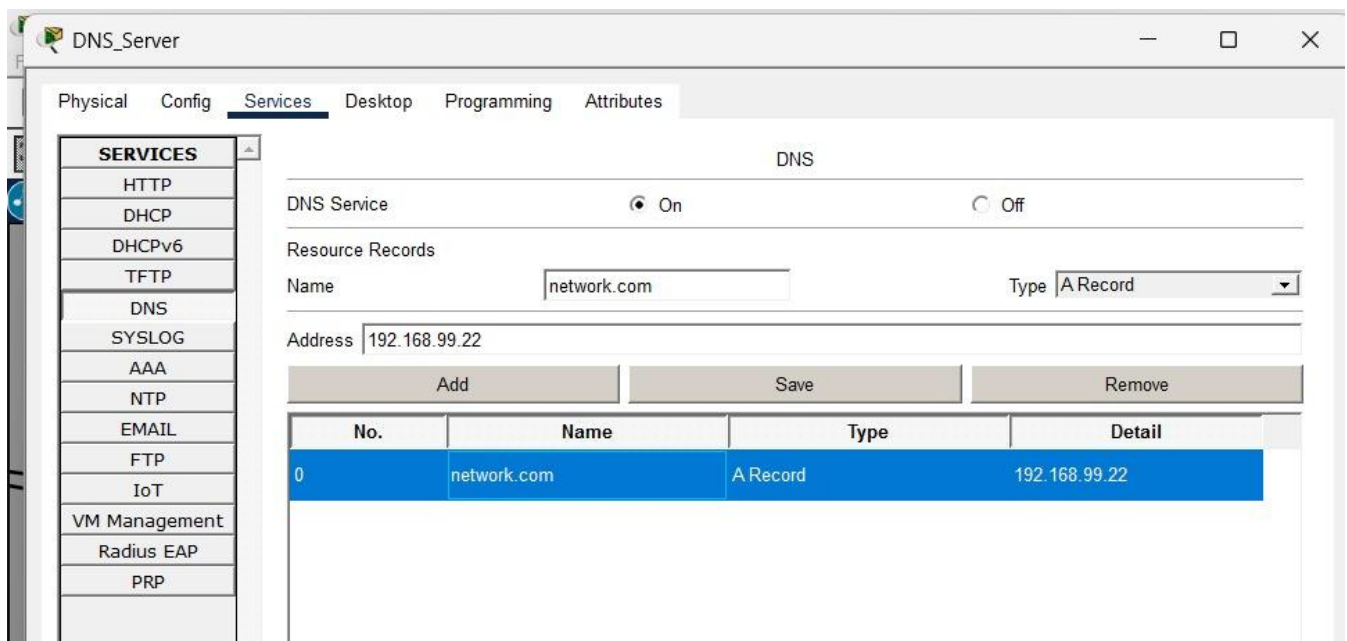
	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

Click on edit of index.html in the File Name → Replace the code →
Click on Save



DNS Configuration:

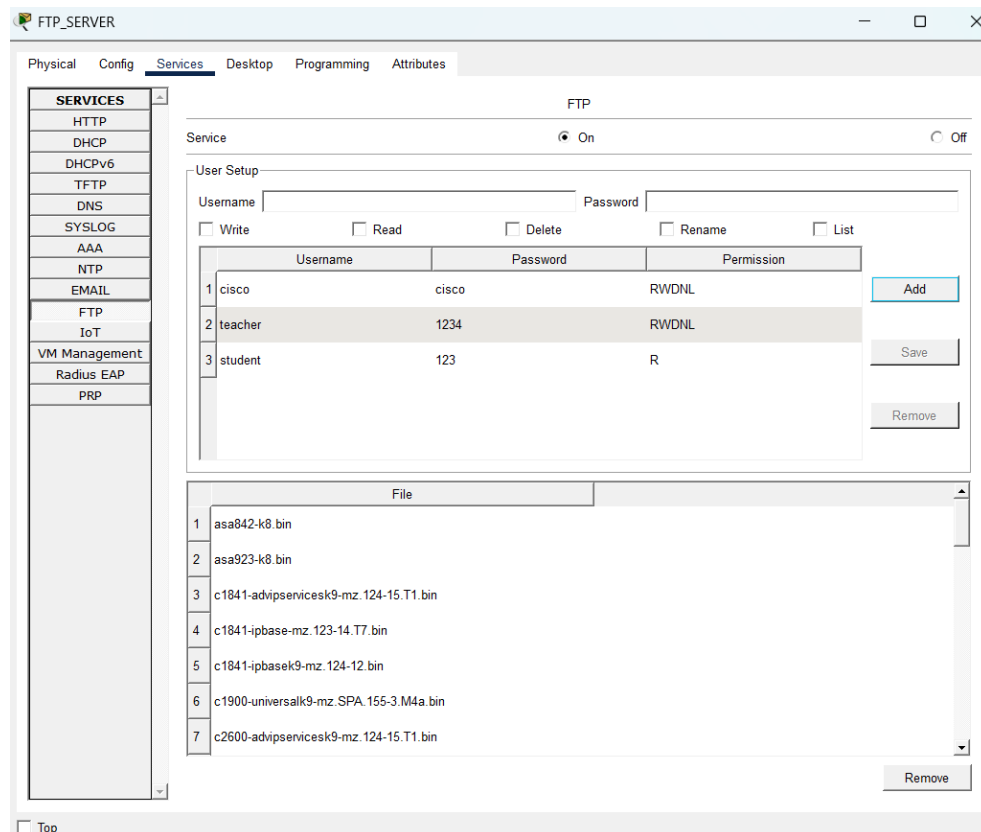
Click on DNS Server → Go to Services → Select DNS → Turn on
DNS → Enter the html file name and IP address of WEB Server →
Add the file



FTP Configuration:

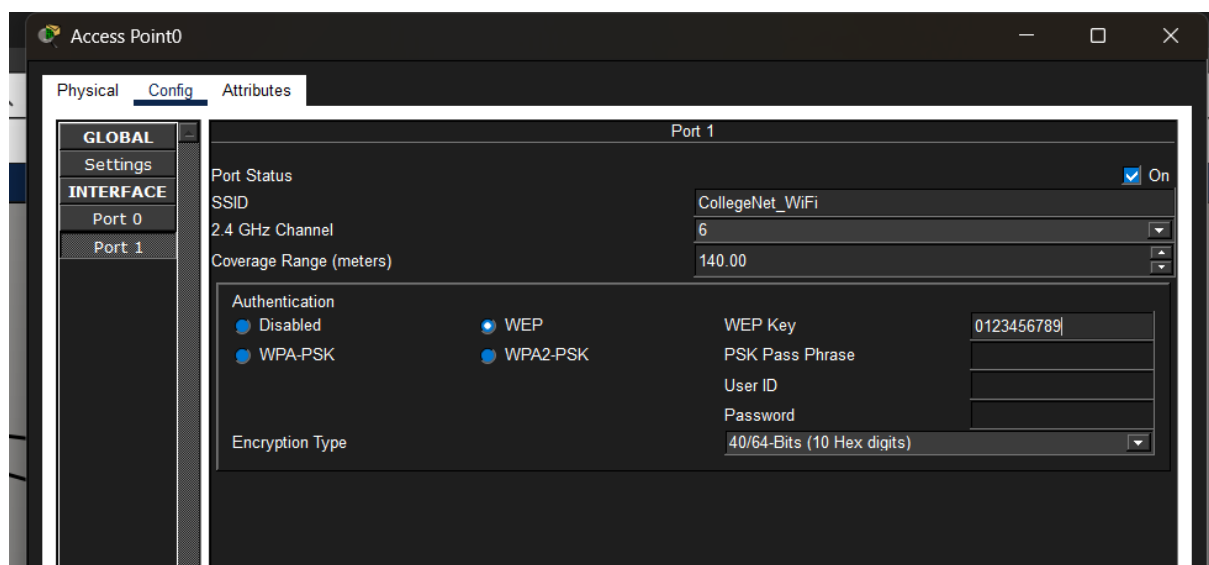
Click on FTP Server → Go to Services → Select FTP → Turn on FTP

Write name and put password → Give permissions → Click on Add



Access Point Configuration:

Click on Access Point → Go to Config → Select Global → Click on Port1 and Change SSID → Click on WEP → Set WEP Key



Output:

DHCP Verification

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.99.21

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:D3FF:FE13:5E69

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC9

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.40.14

Subnet Mask 255.255.255.0

Default Gateway 192.168.40.1

DNS Server 192.168.99.21

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:5CFF:FE73:7E08

Default Gateway

DNS Server

802.1X

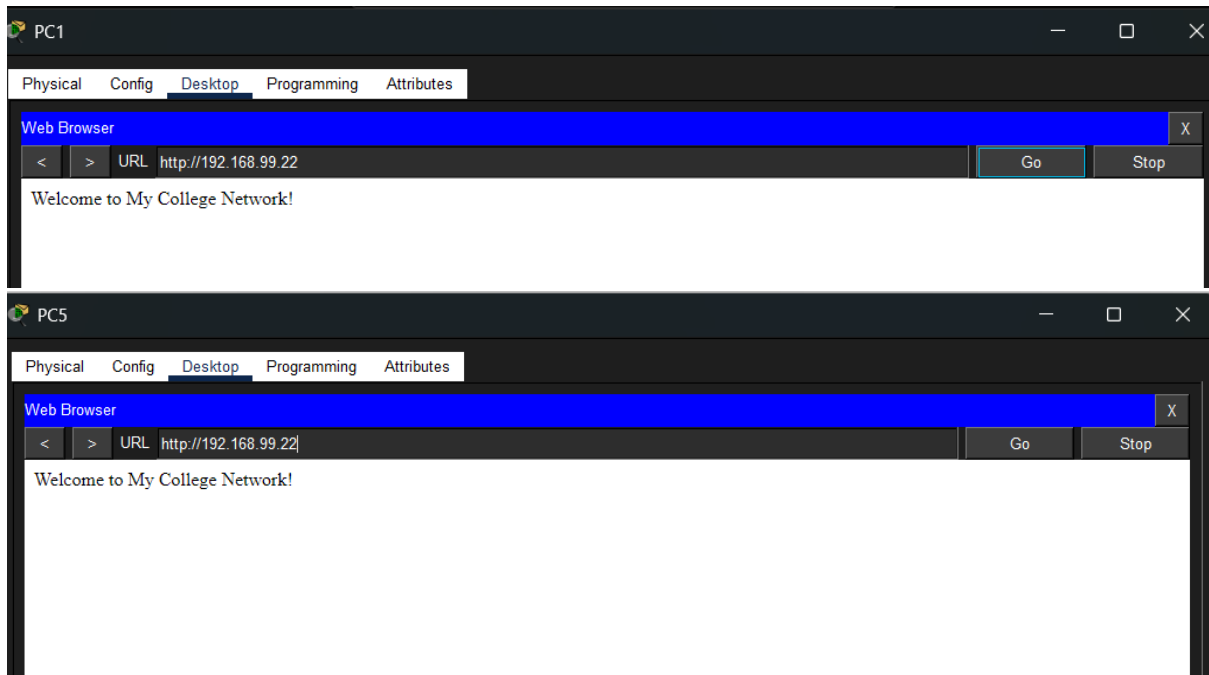
☐ Use 802.1X Security

Authentication MD5

Username

Password

WEB Server Verification



DNS Verification



FTP Verification

```
PC3
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.99.23
Trying to connect...192.168.99.23
Connected to 192.168.99.23
220- Welcome to FT Ftp server
Username:teacher
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.99.23:
 0 : asa842-k8.bin                    5571584
 1 : asa923-k8.bin                    30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ipbase-mz.123-14.T7.bin    13832032
 4 : c1841-ipbasek9-mz.124-12.bin     16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin            5571584
 8 : c2600-ipbasek9-mz.124-8.bin      13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4a.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12 : c2800nm-ipbasek9-mz.124-8.bin    15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-i6q412-mz.121-22.EA4.bin  3058048
15 : c2950-i6q412-mz.121-22.EA8.bin  3117390
```

```
PC3
Physical Config Desktop Programming Attributes
Command Prompt X
31 : ir800-yocto-1.7.2.python-2.7.9.tar 8912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin            5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin  3117390
ftp>put cn.txt

Writing file cn.txt to 192.168.99.23:
File transfer in progress...

[Transfer complete - 6 bytes]

6 bytes copied in 0.04 secs (150 bytes/sec)
ftp>dir

Listing /ftp directory from 192.168.99.23:
 0 : asa842-k8.bin                    5571584
 1 : asa923-k8.bin                    30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ipbase-mz.123-14.T7.bin    13832032
 4 : c1841-ipbasek9-mz.124-12.bin     16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin            5571584
```

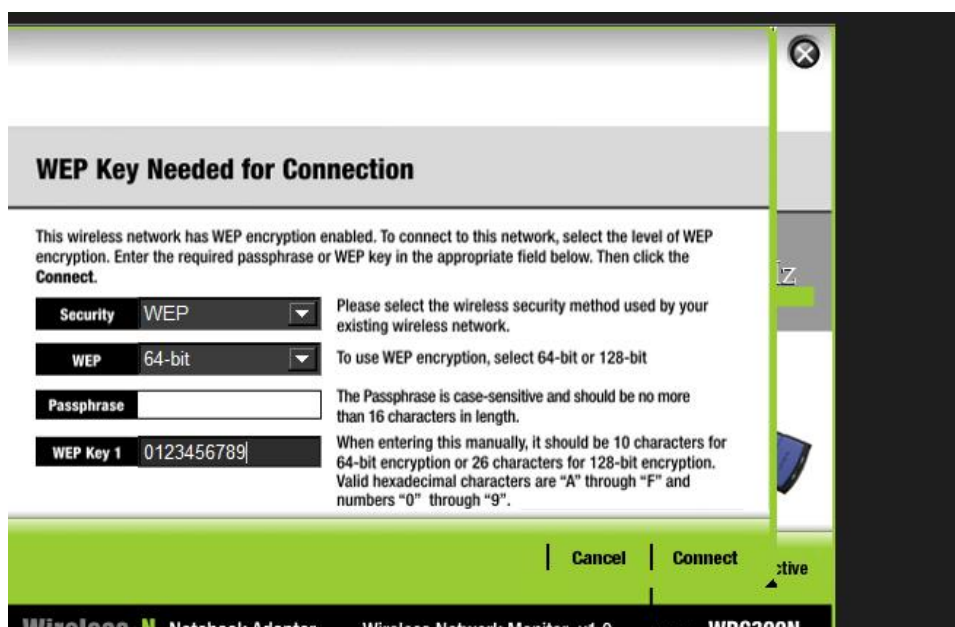
```
17 : c2960-lanbase-mz.122-25.SEE1.bin  4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin  4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE1.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4a.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : cn.txt                             6
27 : ie9k_iosxe.17.09.04.SPA.bin       596133776
28 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
29 : ir800-universalk9-mz.SPA.155-3.M  61750062
30 : ir800-universalk9-mz.SPA.156-3.M  63753767
```

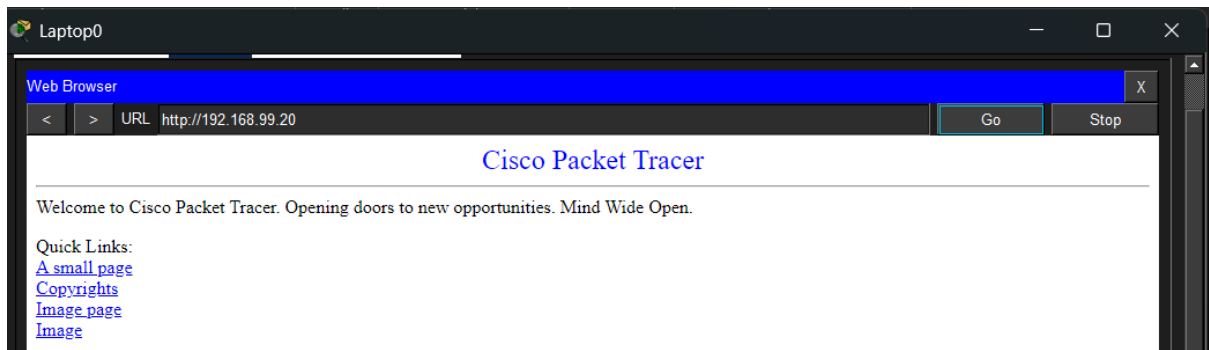
```
PC10
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.99.23
Trying to connect...192.168.99.23
Connected to 192.168.99.23
220- Welcome to FT Ftp server
Username:student
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.99.23:
%Error ftp://192.168.99.23/ (No such file or directory Or Permission denied)
550-Requested action not taken. permission denied).

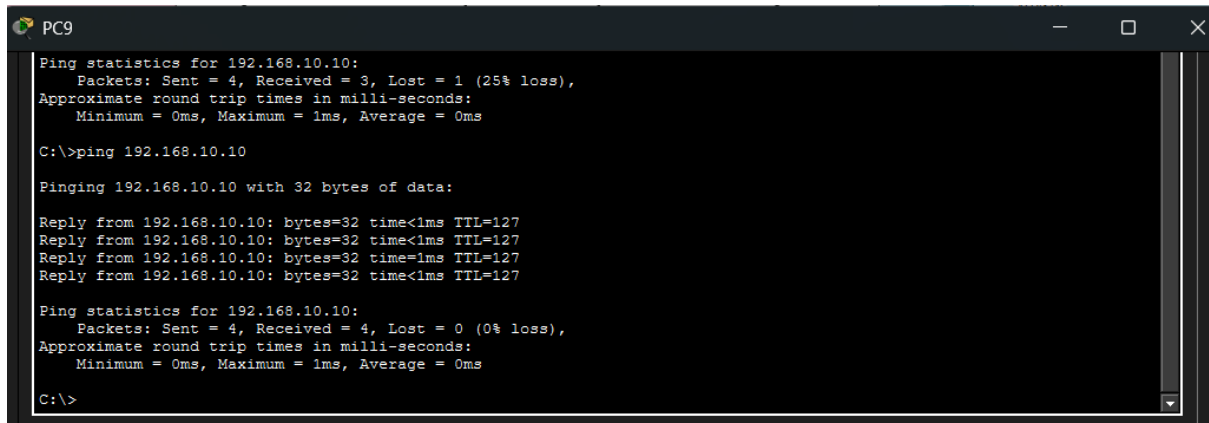
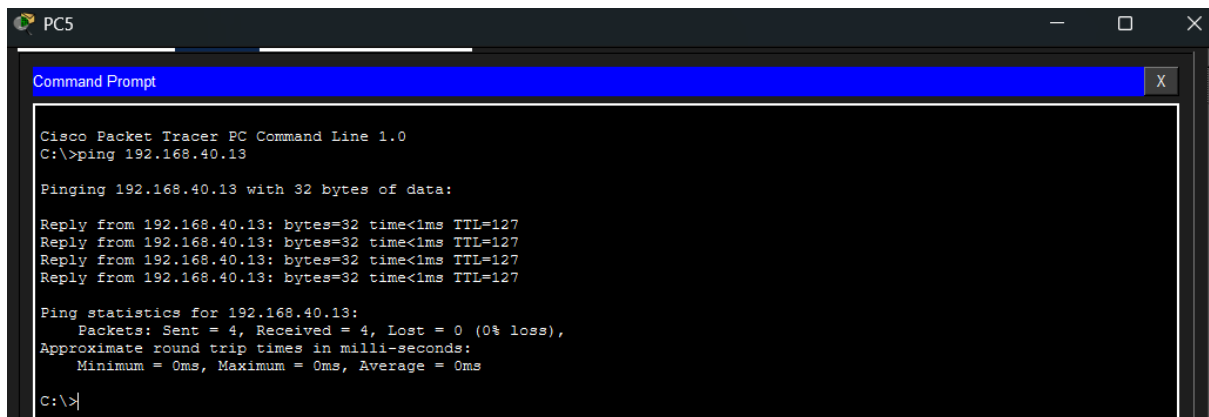
ftp>
```

Access Point Verification





Inter VLAN Verification



Conclusion:

The College Network System was successfully designed and implemented using Cisco Packet Tracer. The project achieved reliable communication between different departments while maintaining an organized network structure through VLANs. Network services such as DHCP, DNS, Web Server, FTP Server, and Wireless Access Point were configured to provide efficient resource sharing and connectivity. The project also demonstrated basic network security using passwords, VLAN segmentation, and user access permissions. Overall, this implementation helped in understanding the practical concepts of computer networking, routing, switching, and network management. The network can be further enhanced in the future by adding advanced security features such as firewalls, ACLs, and network monitoring tools.